

TY B. Tech Artificial Intelligence and Data Science
Pattern 2023 Semester: VI
2301315A: Cloud Computing

Teaching Scheme:	Credit Scheme:	Examination Scheme:
Theory: 03 hrs/week	03	Continuous Comprehensive Evaluation: 20 Marks InSem Exam: 20 Marks EndSem Exam: 60 Marks

Prerequisite Courses: - COM223009 Data Communications & Networking

Course Objectives:

- To understand the concepts of Cloud Computing.
- To learn Taxonomy of Virtualization Techniques.
- To learn Cloud Computing Architecture.
- To acquire knowledge on Aneka Cloud Application Platform.
- To learn Industry Cloud Platforms.

Course Outcomes: On completion of the course, students will be able to –

	Course Outcomes	Bloom's Level
CO1	Explain the fundamental concepts and architecture of cloud computing	2-Understand
CO2	Describe various enterprise and cloud storage systems and their role in distributed data management.	2-Understand
CO3	Summarize the concepts of virtualization, its types, architecture, and its integration with cloud computing infrastructure.	2-Understand
CO4	Illustrate how cloud platforms like AWS and Google Cloud support real-world applications in fields	2-Understand
CO5	Discuss the risks, challenges, and security measures related to data confidentiality, integrity, and availability in the cloud.	2-Understand

COURSE CONTENTS

Unit I	Introduction	(06 hrs)	CO1
Importance of Cloud Computing, Characteristics, Pros and Cons of Cloud Computing, Migrating into the Cloud, Seven-step model of migration into a Cloud, Trends in Computing. Cloud Service Models: SaaS, PaaS, IaaS, Storage. Cloud Architecture: Cloud Computing Logical Architecture, Developing Holistic Cloud Computing Reference Model, Cloud System Architecture, Cloud Deployment Models			
Unit II	Data Storage and Cloud Computing	(06 hrs)	CO2
Data Storage: Introduction to Enterprise Data Storage, Direct Attached Storage, Storage Area Network, Network Attached Storage, Data Storage Management, File System, Cloud Data Stores, Using Grids for Data Storage. Cloud Storage: Data Management, Provisioning Cloud storage, Data Intensive Technologies for Cloud Computing. Cloud file system (gfs and hdfs) Cloud Storage from LANs to WANs: Cloud Characteristics, Distributed Data Storage			
Unit III	Virtualization in Cloud Computing	(08hrs)	CO3

Introduction: Definition of Virtualization, Adopting Virtualization, Types of Virtualization, Virtualization Architecture and Software, Virtual Clustering, Virtualization Application, Pitfalls of Virtualization. Grid, Cloud and Virtualization: Virtualization in Grid, Virtualization in Cloud, Virtualization and Cloud Security. Virtualization and Cloud Computing: Anatomy of Cloud Infrastructure, Virtual infrastructures, CPU Virtualization, Network and Storage Virtualization.			
Unit IV	Cloud Platforms and Cloud Applications	(08hrs)	CO4
Amazon Web Services (AWS): Amazon Web Services and Components, Amazon Simple DB, Elastic Cloud Computing (EC2), Amazon Storage System, Amazon Database services (Dynamo DB). Cloud Computing Applications: Healthcare: ECG Analysis in the Cloud, Biology: Protein Structure Prediction, Geosciences: Satellite Image Processing, Business and Consumer Applications: CRM and ERP, Social Networking, Google Cloud Application: Google App Engine. Overview of OpenStack architecture.			
Unit V	Security in Cloud Computing	(08hrs)	CO5
Risks in Cloud Computing: Risk Management, Enterprise-Wide Risk Management, Types of Risks in Cloud Computing. Data Security in Cloud: Security Issues, Challenges, advantages, Disadvantages, Cloud Digital persona and Data security, Content Level Security. Cloud Security Services: Confidentiality, Integrity and Availability, Security Authorization Challenges in the Cloud, Secure Cloud Software Requirements, Secure Cloud Software Testing.			
Text Books			
1. A. Srinivasan, J. Suresh, "Cloud Computing: A Practical Approach for Learning and Implementation", Pearson, ISBN: 978-81-317-7651-3 2. Gautam Shrof, "ENTERPRISE CLOUD COMPUTING Technology Architecture, Applications", Cambridge University Press, ISBN: 9780511778476			
Reference Books			
1. Rajkumar Buyya, Christian Vecchiola, S. Thamarai Selvi, "Mastering Cloud Computing", 2. Dr. Kris Jamsa, "Cloud Computing: SaaS, PaaS, IaaS, Virtualization and more", Wiley Publications, ISBN: 978-0-470-97389-9 3. Tim Mather, Subra K, Shahid L., "Cloud Security and Privacy", Oreilly, ISBN-13 978-81-8404-815-5 4. Dr. Kumar Saurabh, "Cloud Computing, 4ed: Architecting Next-Gen Transformation Paradigms", Wiley publication, ISBN: 9788126570966 5. Rishabh Sharma, "Cloud Computing: Fundamentals, Industry Approach and Trends", Wiley publication			

Strength of CO-PO PSO Mapping														
	PO												PSO	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO1	3	2	3	2	-	-	-	-	3	2	-	3	3	2
CO2	3	3	2	2	-	-	-	-	3	2	-	3	3	3
CO3	3	3	3	2	-	-	-	-	3	2	-	3	3	3
CO4	3	3	3	2	-	-	-	-	3	2	-	3	3	3
CO5	3	3	3	2	-	-	-	-	3	2	-	3	3	3
Average	3	3	2	2	-	-	-	-	3	2	-	3	3	3

Guidelines for Continuous Comprehensive Evaluation of Theory Course		
Sr. No.	Components for Continuous Comprehensive Evaluation	Marks Allotted
1	Unit I - Quiz (10 marks and will be converted to 4 marks)	4
2	Unit II - Quiz (10 marks will be converted to 4 marks)	4
3	Unit III - Written Test (10 marks will be converted to 4 marks)	4
4	Unit IV - Written Test(10 marks and will be converted to 4 marks)	4
5	Unit V - Presentation (10 marks and will be converted to 4 marks)	4
Total		20